

PYTHON CO1 AT2 ANALYTICAL PROBLEM SOVING

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Student Submissions — Analytical

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Q1. Harshad number

ANSWER

```
n = int(input())
s = 0
temp = n
while temp > 0:
    s += temp % 10
    temp //= 10
if n % s == 0:
    print(n, "is a Harshad number")
else:
    print(n, "is not a Harshad number")
for i in range(1, 501):
    total = 0:
    t = i
    while t > 0:
        total += t % 10
        t //= 10
    if i % total == 0:
        print(i, end=" ")

input
18

output
18 is a Harshad number
1 2 3 4 5 6 7 8 9 10 12 18 20 21 24 27 30 ....
```

Q2. GCD

CODE

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))
while b != 0:
    temp = a
    a = b
    b = temp % b
print("GCD is", a)
```

OUTPUT

Enter first: Enter second: GCD is 6

Q3. sum to a target value

ANSWER

```
1. set()
2. lst
3. num
4. seen
5. seen
6. pairs
7. find_pairs
```

Q4. Bitwise Operators

ANSWER

```
a=13, b=21, c=26, result=31
0b11111 0o37 0x1f
7 62 -32

DATA TYPES
```

Q4. Bitwise Operators

ANSWER

```
a=13, b=21, c=26, result=31
0b11111 0o37 0x1f
7 62 -32
```

DATA TYPES

```
a: int
b: int
c: input
result: int
```

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Q5. binary search

CODE

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        if arr[low] <= arr[mid]:
            if target >= arr[low] and target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if target > arr[mid] and target <= arr[high]:
                low = mid + 1
            else:
                high = mid - 1
    return -1
data = [4, 5, 6, 7, 0, 1, 2]
t = int(input())
result = rotated_binary_search(data, t)
print("Found at:", result)
```

OUTPUT

Found at: 4